DSA REFERENCE

Union-Find Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Maintain disjoint components under repeated merges with parent links, path compression, and union by rank or size.

VERSION 1.0 REVIEW

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STATUS HUMAN REVIEW REQUIRED

DSA / UNION-FIND

DSA

Union-Find Failure Modes

CORE IDEA

Maintain disjoint components under repeated merges with parent links, path compression, and union by rank or size.

Recognition cues

+ dynamic connectivity
+ redundant edge
+ component count after unions
+ Kruskal minimum spanning tree

Implementation sequence

+ initialize each element as its own root
+ compress during find
+ merge roots not raw nodes
+ update count only on a real merge

Invariant

Every element reaches exactly one representative; component metadata belongs only to representatives.

Worked reasoning

An edge whose endpoints already share a representative is redundant and closes a cycle.

Complexity

Nearly O(1) amortized per operation: O(alpha(n)); O(n) storage.

Failure modes

- [ ] merging non-root nodes
- [ ] incorrect rank updates
- [ ] counting duplicate unions
- [ ] using it when deletions or path details are required

Practice ladder

Union-Find Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Number of Provinces**	State the cue, invariant, edge case, and proof before implementation.
2	**Redundant Connection**	State the cue, invariant, edge case, and proof before implementation.
3	**Accounts Merge**	State the cue, invariant, edge case, and proof before implementation.
4	**Graph Valid Tree**	State the cue, invariant, edge case, and proof before implementation.
5	**Most Stones Removed**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
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